# Omar Elarabi

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**LinkedIn** **Portfolio**

## EDUCATION

**Queen Mary University of London, Aerospace Engineering, 1st Class Honours** **September 2023 - July 2026**

- ***Key Modules Completed:*** Designing for Sustainable Manufacture, Control System Analysis and Design, Flight Conceptual Design, Materials Engineering, Aircraft Propulsion Systems & Structures, Aerothermodynamics of Flows, High-Speed Flows.

## PROFESSIONAL EXPERIENCE

**Peer-Assisted Study Support (PASS) Student Organiser** **August 2025 - June 2026**

- Monitored session schedules and room allocations using Excel-based analysis, checklists of essential requirements, and coordination with departments, whilst planning promotional activities and coordinating 10 mentors to deliver sessions.
- Maintained module page and feedback analytics, liaised with staff and professors to align sessions with student needs, and implemented improvements based on data-driven insights, increasing engagement by 40%.

**Peer-Assisted Study Support (PASS) Mentor** **September 2024 - June 2026**

- Recognised and addressed specific areas where students struggled, developing strategies to address these weaknesses and improve overall performance by an average of 60%.
- Customised teaching approaches and managed sessions based on individual student needs, fostering a supportive learning environment and driving academic progress. Awarded Mentor of the Year.

## ACADEMIC PROJECTS

**'Designing the Powertrain Exhaust System – Formula Student UK' – SolidWorks & ANSYS Fluent** **September 2025 - June 2026**

- Researched sustainable materials and manufacturing processes to optimise performance, weight, and sustainability. Designed headers, collector, tailpipe, and muffler in CAD, ensuring fit and compliance with chassis and IMechE rules. Used CFD and thermal simulations to refine gas flow, scavenging, and heat management. Conducted lifecycle analysis to guide material selection and sustainable manufacturing. Validated performance, noise compliance, and compatibility with vehicle.

**'Redesigning the A320 Flap Track Fairing – National Composites Centre' – CAD & ANSYS** **February 2025 - March 2025**

- Screened over 4,000 materials in ANSYS EduPack and applied Ashby's method to select a composite for the A320 flap track fairing, meeting constraints of 3.4 kg mass, 1.66 mm deflection, and stability at 44,000 ft. Refined the material composition to maximise stiffness-to-density ratio and sustainability, defining prepreg layup and autoclave curing methods. Incorporated sustainable strategies such as downcycling analysis and precision cutting to minimise waste. Presented to NCC stakeholders.

**'Artificial Intelligence in Air Traffic Management – New Zealand' – Python & MATLAB** **April 2024 - May 2024**

- Constructed an AI machine learning model using Python and MATLAB, to predict taxi time within a provided airport using a PCA and generating a fuzzy inference system, reducing 37 features of over 2000 provided flight dataset, to 29. Categorised the features into 4 principal components, depending on significance of features in training and testing the model. Removed outliers and anomalies in the data to create an accurate predictive model with a low RMSE, receiving a first-class grade.

## EXTRA-CURRICULAR & LEADERSHIP ROLES

**Aurora Initiative, Health & Safety Officer, QMUL** **August 2025 - June 2026**

- Conducted safety inductions and pre-activity briefings for society members, developed accessible briefing materials, and led annual risk assessments to mitigate hazards across all events and engineering activities.
- Maintained compliance and attendance records, coordinated with other officers and departments to update safety protocols, and implemented preventative measures that ensured activities ran safely and in line with engineering standards.

**Hyperlink, Braking Team Member, Queen Mary University of London** **September 2024 - May 2025**

- Designed concepts using sketching and CAD, of magnetic braking methods for primary and secondary braking systems including use of eddy currents for magnetic damping and braking pads for frictional drag.
- Incorporated dimensions and material research into maximising efficiency for primary system, and methods to primary system, whilst considering health and safety, including power faults.

**Second Year Aerospace Engineering Course Representative, QMUL** **September 2024 - May 2025**

- Built a communication channel between a sum of 190 students and staff to improve student experience in partnership with QMUL, students and student unions.
- Raised awareness of issues in meetings, reporting feedback from meetings and recording turnouts after improvements.

## SKILLS & INTERESTS

**IT Skills:** Computer-aided Design, MATLAB, Python, Abaqus, Microsoft Office, Computational Fluid Dynamics, StarCCM+, ANSYS.